



On an isoperimetric problem for Hamming graphs

L.H. Harper

Department of Mathematics, University of California, Riverside, CA 92521, USA

Received 19 January 1998; received in revised form 20 July 1998; accepted 23 December 1998

Abstract

We consider the vertex-isoperimetric problem for the product of complete graphs. This was posed to us by A.R. Urbanke who wished to apply it in error-correcting coding theory. The problem does not have nested solutions, so the known combinatorial methods are ineffectual. However, passing to the continuous limit gives a variational problem, interesting in its own right, which we solve exactly. Its solution gives a lower bound on the original combinatorial problem which is asymptotically best possible and even sharp in many cases. © 1999 Elsevier Science B.V. All rights reserved.

MSC: Primary 05C35; 05A20; secondary 05C78; 05C90

Keywords: Vertex-isoperimetric problem; Product of complete graphs

1. Introduction

The vertex-isoperimetric problem is a fundamental one in graph theory and its applications: Given a graph, $G=(V,E,\partial)$, having a vertex-set V , edge-set E and function $\partial: E \rightarrow \binom{V}{2}$ which specifies the end-points of each edge, and an integer ℓ , $0 \leq \ell \leq |V|$, minimize $|\Phi(S)|$ over all $S \subseteq V$, $|S| = \ell$, where

$$\Phi(S) = \{w \in V - S: \exists e \in E, \quad \partial(e) = \{v, w\}, \quad v \in S\}$$

is the *vertex-boundary* of S . The vertex-isoperimetric problem is NP-complete [8] in general, so no polynomially bounded solution is known and it is unlikely that one exists. The cases which arise in applications though, often have special structure, like sparseness or symmetry, which facilitate their solution.

R. Urbanke, working in (error correction) coding theory, has asked us about the vertex-isoperimetric problem for $(K_q)^d$, the d -fold product of complete graphs on vertex sets $V_{K_q} = \{0, 1, \dots, q-1\}$. This is also called the *Hamming graph*, since two vertices

E-mail address: harper@math.ucr.edu (L.H. Harper)

have an edge between them iff they are Hamming distance 1 apart (differ in exactly one coordinate) [21]. A combinatorial solution for the case $q = 2$ was presented in one of the author’s first papers [12]. Later that was extended to $(P_q)^d$, the d -fold product of paths on q vertices by Chvatalova ($d = 2$) [5] and Moghadam ($d > 2$) [19] and independently by Bollobas and Leader [2]. Subsequent study of these results and variants [15] has made it clear that their combinatorial methods do not extend to Urbanke’s Problem: The main method, called *compression* (see Appendix B), depends upon the problem having *nested solutions*, $S_0 \subset S_1 \subset \cdots \subset S_{|V|}$, one for each value of $\ell = |S_\ell|$, with $|\Phi(S_\ell)| = \min_{|S|=\ell} |\Phi(S)|$. Urbanke’s Problem for $q > 2$ does not have nested solutions for $d > 1$. A related method, called *stabilization* (see Appendix A), does not apply to it either.

Results on the edge-analog of Urbanke’s Problem also provide perspective: The *edge-isoperimetric problem* on $G=(V,E,\partial)$ is to minimize $|\partial(S)|$ for all $S \subseteq V$, $|S|=\ell$, where

$$\partial(S) = \{e \in E: \partial(e) = \{v,w\}, v \in S \text{ \& \& } w \notin S\}$$

is the *edge-boundary* of S . A combinatorial solution of the edge-isoperimetric problem on the graph of the d -dimensional cube, $(K_2)^d$, was given in [11]. This was quickly followed by its extension to $(K_q)^d$, by Lindsey [18]. However the edge-isoperimetric problem for $(P_q)^d$, the d -dimensional square grid, does not have nested solutions and so the combinatorial methods fail. Only 25 years later did Bollobas and Leader [1] solve the problem using analysis. This author had independently arrived at the same conclusion (that analysis is the way to attack these discrete isoperimetric problems which do not have nested solutions) and applied it to an edge-isoperimetric problem posed by Kleitman and West [14]. The method of [14] is very general, being based on the calculus of variations, and should be a paradigm for solving similar problems (of which there are many possibilities). In order to avoid encumbering our text with general discussion, an appendix on the variational method (Appendix C) has been included.

In his 1991 survey of discrete isoperimetric inequalities [17, p. 71]. Leader wrote “It might seem rather surprising that the best (vertex-)isoperimetric inequality in a product of complete graphs is still not known, as this is one of the most basic graphs of combinatorics.” I myself, under the guidance of Ed Posner at JPL, had looked at the problem about 25 years before Leader’s article, along with the problems later solved by Lindsey and Chvatalova et al. At that time the Lindsey and Chvatalova Problems looked doable; technically difficult but doable. The vertex-isoperimetric problem for the product of complete graphs looked impossibly hard. In our yet-to-be-published monograph summarizing the theory developed for such problems [15], we state that “The difficult is now easy and the formerly impossible merely difficult”. It was just such a problem as this that we had in mind; all that was needed was motivation to do the necessary hard work. Urbanke’s proposed application to coding theory provided that.

2. Main results

2.1. Discovering the solution

Case $d=1$: For $d=1$ the solution of Urbanke's Problem is trivial: For any $S \subseteq V_{K_q}$, if $|S|=\ell$, then $|\Phi(S)|=q-\ell$, so all sets are solutions. In particular, $S_\ell=\{0, 1, \dots, \ell-1\}$ is a solution set and K_q has nested solutions $S_0 \subset S_1 \subset \dots \subset S_q$, with $|S|=\ell$. Moghadam, in his thesis [19] called this *Property A*. Property A is necessary for compression to work on vertex-isoperimetric problems, but there is another condition, which he called *Property B*, which is also necessary:

$$\forall \ell, \quad S_\ell \cup \Phi(S_\ell) = S_{\ell+|\Phi(S_\ell)|}.$$

Our family of solution sets for K_q , also trivially, has Property B. Properties A and B together are sufficient for compression (see Appendix B).

Case $d=2$: By the above remarks, compression applies to $K_q \times K_q = (K_q)^2$, and we may restrict our sets, S , to lower sets (see Appendix B) in the product order, $\mathcal{T}_q \times \mathcal{T}_q$, where $\mathcal{T}_q = \{0 < 1 < \dots < q-1\}$ is the natural total order on V_{K_q} . For $\ell=1$ there is then only one compressed set, $S_1 = \{(0,0)\}$, which, of course, minimizes $|\Phi(S)|$ for its cardinality. But $\Phi(S_1 \cup \Phi(S_1)) = V - (S_1 \cup \Phi(S_1))$ (the largest possible boundary!) and if $q > 2$, $(K_q)^2$ does not have Moghadam's Property B for $\ell=1$.

2.2. Passage to the continuous limit

By the preceding remarks, compression is only applicable to Urbanke's Problem w.r.t. one-dimensional factors (of $(K_q)^d$). This does mean that we can restrict our search for sets minimizing $|\Phi(S)|$ to lower sets, S , of $(\mathcal{T}_q)^d$, but leaves us far short of a solution. So, as we did with the Kleitman–West Problem, we abandon the direct combinatorial approach and bring in analysis: Let $q \rightarrow \infty$ and normalize $\mathcal{T}_q$ (dividing by q). In the limit, $(\mathcal{T}_q/q)^d$ converges to $([0,1])^d$, the d -fold product of closed unit intervals with the product partial order, cardinality converges to Lebesgue measure (which we continue to denote with vertical bars) and for S a lower set in $[0,1]^d$ we define

$$\Phi(S) = \{y \in [0,1]^d - S : \exists x \in S \text{ \& \; } \exists i_0 \text{ s.t. } \forall i \neq i_0, \ y_i = x_i\}.$$

Urbanke's Problem then converges (in an intuitive sense which can be made precise but is not necessary in the ultimate logic of our solution) to

The Continuous Urbanke Problem. Given v , $0 \leq v \leq 1$, minimize $|\Phi(S)|$ over all lower sets, S , of $([0,1])^d$ with $|S|=v$ (we will not distinguish between sets whose symmetric difference has measure 0).

The Continuous Urbanke Problem is a problem of the calculus of variations. The space of all possible solutions is infinite-dimensional, so it may well seem that we

have made negative progress by going from a finite problem to an infinite-dimensional one. However, we are now able to apply variational arguments:

- Any lower set, S , of $([0, 1])^d$, can be approximated arbitrarily closely by a finitely generated lower set: A sequence, $0=t_0 < t_1 < \cdots < t_q=1$, defines an order-morphism $\varphi : [0, 1] \rightarrow \mathcal{T}_q$ by letting $\varphi(x) = j$ if $t_j \leq x < t_{j+1}$. If we have a set of such morphisms, $\{\varphi_i : [0, 1] \rightarrow \mathcal{T}_{q_i}, 1 \leq i \leq d\}$, it defines an order-morphism

$$\varphi = \varphi_1 \times \varphi_2 \times \cdots \times \varphi_q : [0, 1]^d \rightarrow \mathcal{T}_{q_1} \times \mathcal{T}_{q_2} \times \cdots \times \mathcal{T}_{q_d}.$$

Note that $T=\varphi(S)$ is a lower set (of $\mathcal{T}_{q_1} \times \mathcal{T}_{q_2} \times \cdots \times \mathcal{T}_{q_d}$) as is $\varphi^{-1}(T)$. Equivalently, φ is determined by the sequence of sequences

$$\{t_{i,j}: 1 \leq i \leq d \text{ \& } 0 \leq j \leq q_i\}.$$

It then follows from basic properties of Lebesgue measure [10] that, for any $\varepsilon > 0$, there exists a $\delta > 0$, such that if

$$\max\{t_{i,j+1} - t_{i,j}: 1 \leq i \leq d \text{ \& } 0 \leq j < q_i\} < \delta$$

then

$$|S \Delta (\varphi^{-1} \circ \varphi(S))| < \varepsilon.$$

Note that

$$|\varphi^{-1}(T)| = \sum_{x \in T} \prod_{i=1}^d (t_{i,x_i+1} - t_{i,x_i})$$

which we shall denote by $|T; t|$. Also

$$|\Phi(T); t| = \sum_{x \in \Phi(T)} \prod_{i=1}^d (t_{i,x_i+1} - t_{i,x_i}),$$

where

$$\Phi(T) = \{y \in \mathcal{T}_{q_1} \times \mathcal{T}_{q_2} \times \cdots \times \mathcal{T}_{q_d} - T: \exists x \in T \text{ \& } \exists i_0 \text{ s.t. } \forall i \neq i_0, y_i = x_i\}.$$

- If $S = \varphi^{-1}(T)$ is such a finitely generated lower set and $j \neq 0$ or q_i , then the set

$$\{x \in S: x_i = t_{i,j} \text{ \& } \forall \varepsilon > 0, x + \delta_i \varepsilon \notin S\}$$

will be called a *facet* of S (δ_i is the vector all of whose entries are 0, except the i th, which is 1). We may raise one facet by a small amount, say ε , and lower another by a corresponding amount, $-f(\varepsilon)$, so as to maintain the volume. If $S = S(t_{i,j}, t_{k,l})$ is our lower set and $S(t_{i,j} + \varepsilon, t_{k,l} + f(\varepsilon))$ is the altered lower set, then by standard techniques of calculus (see Appendix C), the change in $|\Phi|$ will be

$$\begin{aligned} &| \Phi(S(t_{i,j} + \varepsilon, t_{k,l} + f(\varepsilon))) | - | \Phi(S(t_{i,j}, t_{k,l})) | \\ &= \left\{ \frac{\partial |\Phi|}{\partial t_{i,j}} \frac{\partial |S|}{\partial t_{k,l}} - \frac{\partial |\Phi|}{\partial t_{k,l}} \frac{\partial |S|}{\partial t_{i,j}} \right\} \frac{\varepsilon}{\partial |S| / \partial t_{k,l}} \end{aligned}$$

$$\begin{aligned}
& + \left\{ \frac{\partial|\Phi|}{\partial t_{k,l}} \left(-\frac{\partial^2|S|}{\partial t_{i,j}^2} \left(\frac{\partial|S|}{\partial t_{k,l}} \right)^2 + 2 \frac{\partial^2|S|}{\partial t_{i,j} \partial t_{k,l}} \frac{\partial|S|}{\partial t_{i,j}} \frac{\partial|S|}{\partial t_{k,l}} - \frac{\partial^2|S|}{\partial t_{k,l}^2} \left(\frac{\partial|S|}{\partial t_{i,j}} \right)^2 \right) \right. \\
& + \left. \frac{\partial^2|\Phi|}{\partial t_{i,j}^2} \left(\frac{\partial|S|}{\partial t_{i,j}} \right)^3 - 2 \frac{\partial^2|\Phi|}{\partial t_{i,j} \partial t_{k,l}} \frac{\partial|S|}{\partial t_{i,j}} \left(\frac{\partial|S|}{\partial t_{k,l}} \right)^2 + \frac{\partial^2|\Phi|}{\partial t_{k,l}^2} \left(\frac{\partial|S|}{\partial t_{i,j}} \right)^2 \frac{\partial|S|}{\partial t_{k,l}} \right\} \\
& \times \frac{\varepsilon^2}{2(\partial|S|/\partial t_{k,l})^3} + O(\varepsilon^3)
\end{aligned}$$

- If the coefficient of ε is $\neq 0$, then $S(t_{i,j}, t_{k,l})$ cannot minimize $|\Phi(S)|$ (remember, $t_{i,j} \neq 1$ and $t_{k,l} \neq 0$). That coefficient, known as the *first variation*, must be 0 for an optimum lower set. This is equivalent to

$$\frac{\partial|\Phi|/\partial t_{i,j}}{\partial|S|/\partial t_{i,j}} = \frac{\partial|\Phi|/\partial t_{k,l}}{\partial|S|/\partial t_{k,l}},$$

which may be paraphrased as

The optimality condition for lower sets: The relative marginal boundary of any two facets must be equal.

The variational method thus brings up a key concept for the remainder of the paper, *relative marginal boundary*.

- In order to avoid dealing with the second variation (i.e. the coefficient of ε^2 above), we at this point restrict our attention to pairs of facets which are parallel, i.e. for which $i = k$. Such facets do not intersect so $f(\varepsilon) = c\varepsilon$ (with $c = -\partial|S(t_{i,j}, t_{i,l})|/\partial t_{i,j} / \partial|S(t_{i,j}, t_{i,l})|/\partial t_{i,l}$), and $|\Phi|$ is a polynomial in ε for $-(t_{i,j} - t_{i,j-1}) < \varepsilon < t_{i,j+1} - t_{i,j}$ and $-(t_{i,l} - t_{i,l-1}) < f(\varepsilon) < t_{i,l+1} - t_{i,l}$. Each summand in our representation of $|\Phi(T); t|$ has only one factor whose first subscript is i , so the second and all higher variations will be 0. That is to say that $|\Phi(S(t_{i,j} + \varepsilon, t_{i,l} + f(\varepsilon)))| - |\Phi(S(t_{i,j}, t_{i,l}))|$ is just of first degree in ε . Therefore we may always increase or decrease ε , until one of its bounds is violated, without increasing $|\Phi|$. At that point the number of facets has been decreased. We may continue to do this until the number of facets perpendicular to δ_i reaches 1, i.e. $q_i = 2$. Since $\varepsilon > 0$ was arbitrary, we may restrict Urbanke's Problem to lower sets, T , in the (weighted) Boolean lattice, $(\mathcal{T}_2)^d$. In this case we drop the second subscript and define $t_{i,1} = t_i$.
- For such a lower set, T , there are infinitely many weightings, $t \in [0, 1]^d$, for which $|T; t| = v$. The space of all such weightings is compact, however, and the function we seek to minimize, $|\Phi(T); t|$ is continuous, so there is some $t_0 \in [0, 1]^d$ with $|T; t_0| = v$ and

$$|\Phi(T); t_0| = \min\{|\Phi(T); t| : t \in [0, 1]^d, |T; t| = v\}.$$

In the rest of the paper we may drop t from our notation, writing just $|T| = v$ and $|\Phi(T)|$ with the understanding that $t = t_0$, the optimal weighting for T .

- Thus our reduced continuous Urbanke Problem is: Given $v, 0 \leq v \leq 1$, minimize $|\Phi(T)|$ over all lower sets, T of $(\mathcal{T}_2)^d$, with $|T| = v$, the weight function, t , being the optimal one for T unless otherwise noted.

2.3. Closure and duality

Before resuming the consideration of Urbanke’s Problem for low dimensions, we introduce a couple of simplifying principles. These are fairly standard, so the proofs are just stretched:

- *Closure.* For any graph, $G = (V, E, \partial)$, and $S \subseteq V$,

$$\bar{S} = \{y \in V: (S \cup \{y\}) \cup \Phi(S \cup \{y\}) = S \cup \Phi(S)\}$$

is called the *closure* of S . It is easily seen that $S \subseteq \bar{S}$ and $\bar{\bar{S}} = \bar{S}$. Any S such that $\bar{S} = S$ is called *closed*, so $\bar{S}$ is closed. It is also true that $\bar{S} \cup \Phi(\bar{S}) = S \cup \Phi(S)$. Now if S is a lower set in $[0, 1]^d$ then $S \cup \Phi(S)$ is a lower set and for any lower set $|S|$ is an increasing function of any $t_{i,j}$. If $|\bar{S}| > |S| = v$, then we may decrease $t_{i,j}$ ’s until $|\bar{S}| = v$. But this will decrease $|\bar{S} \cup \Phi(\bar{S})|$ also and make $|\Phi(\bar{S})| < |\Phi(S)|$. Thus we have that *if S is optimal in $[0, 1]^d$, then S is closed*. The same applies to quotients of $[0, 1]^d$.

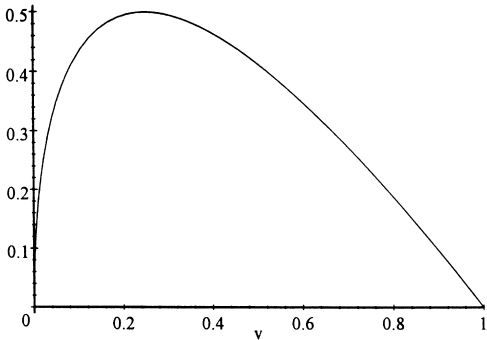
- *Duality.* All of the posets which we have mentioned with respect to Urbanke’s Problem are self-dual. Thus, we may define $DC(S) = (V - (S \cup \Phi(S)))^*$, the dual of the complement of $S \cup \Phi(S)$, and then DC will map lower sets to lower sets. It is easily seen that $DC(DC(S)) = \bar{S}$, so if S is optimal, $DC(DC(S)) = S$. From the definition of $DC(S)$,

$$|S| + |\Phi(S)| + |DC(S)| = 1.$$

We assert then that if S is optimal, $DC(S)$ is also optimal (for its volume) since $|\Phi(DC(S))| = |\Phi(S)|$.

2.4. Discovering the solution (continued from Section 2.1)

Case $d = 2$: The only nontrivial closed lower set in $(\mathcal{T}_2)^2$, is $T_0 = \{(0, 0)\}$. Therefore it must be optimal for all $v, 0 \leq v \leq 1$. $|T_0; (t_1, t_2)| = t_1 t_2$ and $|\Phi(T_0); (t_1, t_2)| = t_1(1 - t_2) + t_2(1 - t_1) = t_1 + t_2 - t_1 t_2$. Thus, given $t_1 t_2 = v$, $|\Phi|$ is minimized with $t_1 = t_2 = \sqrt{v}$.



The graph of $|\Phi(T_0)|(v)$ for $d = 2$

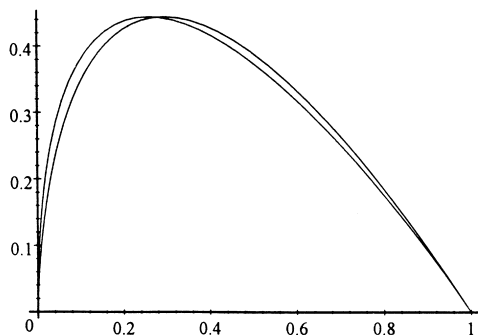
Case $d = 3$: The only candidates for optimality in $(\mathcal{T}_2)^3$ are $T_0 = \{(0, 0, 0)\}$ and $T_1 = DC(T_0) = \{(0, 0, 0), (1, 0, 0), (0, 1, 0), (0, 0, 1)\}$. The set

$$\{(0, 0, 0), (1, 0, 0)\}$$

and its equivalents by permuting the components are essentially the same as T_0 . Also,

$$\{(0, 0, 0), (1, 0, 0), (0, 1, 0)\}$$

and its equivalents are not closed. The rest are dual-complements of these. Again, for our two candidates the best values of t_i , given v , are constant.



Graphs of $|\Phi(T_0)|$ and $|\Phi(T_1)|$ for $d = 3$

So, we see that T_0 is optimal $v \lesssim 0.27$ and T_1 is optimal for $v \gtrsim 0.27$. From such evidence we can guess the following:

Theorem 1. *For every $v, 0 \leq v \leq 1$, the minimum of $|\Phi(T)|$ over all lower sets, $T \subseteq (\mathcal{T}_2)^d$, with $|T| = v$, is achieved by some Hamming ball,*

$$HB(r, d) = \left\{ x \in (\mathcal{T}_2)^d : \sum_{i=1}^d x_i \leq r \right\},$$

$$r = 0, 1, \dots, d - 2.$$

2.5. Proof of Theorem 1

The form of the solutions we have guessed suggests how they might be proven. Since the Hamming balls in $\mathcal{T}_2^d$ are symmetric w.r.t. permutations of the coordinates, and are the only such lower sets, we seek to show that, starting with any lower set, T , in $\mathcal{T}_2^d$, we can make it more and more symmetric while maintaining $|T|$ and not increasing $|\Phi(T)|$. If $\mathfrak{R}_{i,j}$ is the symmetry of $\mathcal{T}_2^d$ defined by interchanging the i th and j th coordinates, then a measure of the asymmetry of T is the symmetric difference of T and $\mathfrak{R}_{i,j}(T)$. If $T \Delta \mathfrak{R}_{i,j}(T) = \emptyset$, then T is symmetric w.r.t. the i th and j th coordinates.

Now, letting $T_a^i = \{x \in T : x_i = a\}$ and $T_{a,b}^{i,j} = (T_a^i)_b^j$, we have

$$T \Delta \mathfrak{R}_{i,j}(T) = [T - \mathfrak{R}_{i,j}(T)] + [\mathfrak{R}_{i,j}(T) - T].$$

Also

$$T - \mathfrak{R}_{i,j}(T) = [T_{0,1}^{i,j} - \mathfrak{R}_{i,j}(T_{1,0}^{i,j})] + [T_{1,0}^{i,j} - \mathfrak{R}_{i,j}(T_{0,1}^{i,j})].$$

If we let $A(T) = T_{0,1}^{i,j} - \mathfrak{R}_{i,j}(T_{1,0}^{i,j})$, $B(T) = T_{1,0}^{i,j} - \mathfrak{R}_{i,j}(T_{0,1}^{i,j})$, then $T \Delta \mathfrak{R}_{i,j}(T) = A + \mathfrak{R}_{i,j}(A) + B + \mathfrak{R}_{i,j}(B)$.

Lemma 1. $T - A(T)$ and $T + \mathfrak{R}_{i,j}(A(T))$ are both lower sets.

Proof. $y \in T - A(T)$ & $x < y \Rightarrow \mathfrak{R}_{i,j}(y) \in T$ & $\mathfrak{R}_{i,j}(x) < \mathfrak{R}_{i,j}(y) \Rightarrow \mathfrak{R}_{i,j}(x) \in T \Rightarrow x \notin A(T)$. The argument for $T + \mathfrak{R}_{i,j}(A(T))$ is similar. $\square$

We now commence with the proof of Theorem 1, defining an operation $SSym_{i,j}$ (called (*semisymmetrization*)) which eliminates $A = A(T)$ and $\mathfrak{R}_{i,j}(A)$. Its definition requires consideration of several cases:

Case 0. $A = \emptyset$: This means we are already done and

$$SSym_{i,j}(T) = T.$$

Case 1.

$$\frac{\partial(|\Phi(T)_0^i| - |A \cup \Phi(T - A)_0^i|)/\partial t_i}{\partial |A|/\partial t_i} > \frac{\partial |\Phi(T)|/\partial t_i}{\partial |T|/\partial t_i}.$$

We will show later (Lemma 4) that the quantity on the left-hand side of this inequality is the relative marginal boundary of that part of the facet orthogonal to δ_i which bounds $\varphi^{-1}(A; t)$. With this in mind, we momentarily reverse some of the simplifying we did in Section 2.2. Consider $S = \varphi^{-1}(T; t)$, a lower set in $[0, 1]^d$. We split t_i into $t_{i,1} = t_i$ (on $\varphi^{-1}(A; t)$) and $t_{i,2} = t_i$ (on $\varphi^{-1}(T - A; t)$). Because of Lemma 1, we may decrease $t_{i,1}$ (add $\varepsilon < 0$) and increase $t_{i,2}$ (add $f(\varepsilon) > 0$) and the set will still be a lower set. Also its volume will be maintained at v until either

- (i) $t_{i,1} = 0$ & $t_{i,2} < 1$: and then we let $SSym_{i,j}(T) = T - A$, or
- (ii) $t_{i,1} > 0$ & $t_{i,2} = 1$: and then we pass to Case 2 with T replaced by

$$T \cup \{x + \delta_i : x \in T_0^i - A\},$$

or

- (iii) $t_{i,1} = 0$ & $t_{i,2} = 1$: and then we let

$$SSym_{i,j}(T) = T - A \cup \{x + \delta_i : x \in T_0^i - A\}.$$

Case 2:

$$\frac{\partial(|\Phi(T)_0^i| - |A \cup \Phi(T - A)_0^i|)/\partial t_i}{\partial |A|/\partial t_i} \leq \frac{\partial |\Phi(T)|/\partial t_i}{\partial |T|/\partial t_i}.$$

Claim 1. *This implies that*

$$\frac{\partial(|\Phi(T + \mathfrak{R}_{i,j}(A))_0^j| - |\mathfrak{R}_{i,j}(A) \cup \Phi(T)_0^j|)/\partial t_j}{\partial |\mathfrak{R}_{i,j}(A)|/\partial t_j} \leq \frac{\partial |\Phi(T)|/\partial t_j}{\partial |T|/\partial t_j},$$

the quantity on the left-hand side being the relative marginal boundary of $\mathfrak{R}_{i,j}(A)$ w.r.t. $T + \mathfrak{R}_{i,j}(A)$.

Assuming this, we consider $S = \varphi^{-1}(T; t)$ as in Case 1, splitting t_j into $t_{j,1} = 0$ (on $\varphi^{-1}(\mathfrak{R}_{i,j}(A); t)$) and $t_{j,2} = t_j$ (on $\varphi^{-1}(T; t)$). Then increase $t_{j,1}$ and decrease $t_{j,2}$ so that the volume of the lower set is maintained at v until $t_{j,1} = t_{j,2}$ and then let

$$SSym_{i,j}(T) = T + \mathfrak{R}_{i,j}(A).$$

This construction gives the following basic properties for $SSym_{i,j}$:

1. $|SSym_{i,j}(T)| = |T|$, by definition,
2. $|\Phi(SSym_{i,j}(T))| \leq |\Phi(T)|$, since (in Case 1(i), and similarly in the other cases)

$$\begin{aligned} |\Phi(T)| &= |\Phi(S(t_{i,1}, t_{i,2}))| \\ &> |\Phi(S(t_{i,1}, t_{i,2}))| + \left(\frac{\partial |\Phi|/\partial t_{i,1}}{\partial |S|/\partial t_{i,1}} - \frac{\partial |\Phi|/\partial t_{i,2}}{\partial |S|/\partial t_{i,2}} \right) \frac{\partial |S|}{\partial t_{i,1}} \varepsilon \\ &\geq |\Phi(SSym_{i,j}(T))|, \end{aligned}$$

because $\varepsilon > 0$ and

Claim 2.

$$\frac{\partial |\Phi|/\partial t_{i,1}}{\partial |S|/\partial t_{i,1}} > \frac{\partial |\Phi|/\partial t_{i,2}}{\partial |S|/\partial t_{i,2}}.$$

Further, $T - A = SSym_{i,j}(T)$ is the set corresponding to $\varepsilon = -t_i$.

3. $A(Sym_{i,j}(T)) = \emptyset$ and $B(SSym_{i,j}(T)) = B(T)$.

We may then define the *symmetrization operator*,

$$Sym_{i,j}(T) = SSym_{i,j}(T) \circ SSym_{j,i}(T)$$

and from the above it follows that

1. $|Sym_{i,j}(T)| = |T|$,
2. $|\Phi(Sym_{i,j}(T))| \leq |\Phi(T)|$,
3. $A(Sym_{i,j}(T)) = \emptyset = B(Sym_{i,j}(T))$, i.e. $Sym_{i,j}(T)$ is symmetric w.r.t. the i th and j th coordinates.

We only need those symmetrization operators with $j = i + 1$, $i = 1, 2, \dots, d - 1$: Define operators S_i , $i = 0, 1, \dots$, recursively by having S_0 be the identity and

$$S_{i+1} = Sym_{i(\bmod (d-1))+1, i(\bmod (d-1))+2} \circ S_i \quad \text{for } i > 0.$$

If $S_i(T)$ is unchanged through a complete cycle of $d - 1$ consecutive symmetrizations, then it will remain the same thereafter and by the Pigeon-Hole Principle this must

happen before i reaches $(d-1)2^d+1$. The reason is that the successive $S_i(T)$ either has a strictly lower value of $|\Phi|$ than its predecessor, or, if $|\Phi(S_{i+1}(T))|=|\Phi(S_i(T))|$ but $S_{i+1}(T)\neq S_i(T)$ (this only happens in Case 2), then $S_i(T)\subset S_{i+1}(T)$. In any case, no lower set of $\mathcal{T}_2^d$ can reappear in the sequence $S_i(T)$, $i=0,1,\dots$ once it has disappeared, and then it can appear no more than $d-2$ times (consecutively) without becoming permanent. Thus $S_\infty(T)$ is well defined and symmetric w.r.t. all permutations of coordinates and must be a Hamming ball of some radius. Since $|S_\infty(T)|=|T|=v$ and $|\Phi(S_\infty(T))|\leq|\Phi(T)|$ we are done, except for Claims 1 and 2, that is.

Proof of Claim 1. We have now arrived at the heart of our argument. The proof of Claim 1 is based on the following lemma borrowed from stabilization theory (cf. Theorem A.1.2 of Appendix A):

Lemma 2. $\mathfrak{R}_{i,j}:\mathfrak{T}_2^d\rightarrow\mathfrak{T}_2^d$, which is one-to-one, takes $\Phi(T+\mathfrak{R}_{i,j}(A))-\Phi(T)$ into $\Phi(T)-\Phi(T-A)$ (remember, $A=A(T)$).

Proof. $y\in\Phi(T+\mathfrak{R}_{i,j}(A))-\Phi(T)$ and y covers $x\Rightarrow x\notin T$ & $(\mathfrak{R}_{i,j}(x)\in T\Rightarrow\mathfrak{R}_{i,j}(x)\in A)$. Also, $\exists x_0$ covered by y such that $\mathfrak{R}_{i,j}(x_0)\in A$. But then $\mathfrak{R}_{i,j}(x_0)$ is covered by $\mathfrak{R}_{i,j}(y)$, so $\mathfrak{R}_{i,j}(y)\in\Phi(T)-\Phi(T-A)$. $\square$

Now we have

$$\begin{aligned} &\frac{\partial(|\Phi(T+\mathfrak{R}_{i,j}(A))_0^j|-|\mathfrak{R}_{i,j}(A)\cup\Phi(T)_0^j|)/\partial t_j}{\partial|\mathfrak{R}_{i,j}(A)|/\partial t_j} \\ &= \frac{|\Phi(T+\mathfrak{R}_{i,j}(A))_0^j|-|\mathfrak{R}_{i,j}(A)\cup\Phi(T)_0^j|}{|\mathfrak{R}_{i,j}(A)|} \\ &\quad \text{since every term of } |\Phi(T)_0^j|, |\mathfrak{R}_{i,j}(A)| \text{ and } |\Phi(T+\mathfrak{R}_{i,j}(A))_0^j| \text{ is of first} \\ &\quad \text{degree in } t_j, \\ &= \frac{|\Phi(T+\mathfrak{R}_{i,j}(A))_0^j-\Phi(T)_0^j|-|\mathfrak{R}_{i,j}(A)|}{|\mathfrak{R}_{i,j}(A)|} \\ &\quad \text{since } |\mathfrak{R}_{i,j}(A)\cup\Phi(T)_0^j|=|\mathfrak{R}_{i,j}(A)|+|\Phi(T)_0^j-\mathfrak{R}_{i,j}(A)| \\ &\quad \text{and } |\Phi(T+\mathfrak{R}_{i,j}(A))_0^j|=|\Phi(T+\mathfrak{R}_{i,j}(A))_0^j-\Phi(T)_0^j|+|\Phi(T)_0^j-\mathfrak{R}_{i,j}(A)|, \\ &= \frac{\sum_{y\in\Phi(T+\mathfrak{R}_{i,j}(A))_0^j-\Phi(T)_0^j}(1-t_i)t_j|y^{(i,j)}|}{\sum_{x\in\mathfrak{R}_{i,j}(A)}(1-t_i)t_j|x^{(i,j)}|}-1, \\ &\quad \text{where for } x\in[0,1]^d, \text{ we let } x^{(i_1,\dots,i_k)}\in[0,1]^{d-k} \text{ be the vector} \\ &\quad \text{obtained from } x \text{ by eliminating its components } i_1,\dots,i_k, \end{aligned}$$

$$\begin{aligned}
&\leq \frac{\sum_{y \in \Phi(T)_0^i - \Phi(T-A)_0^i} t_i(1-t_j)|y^{(i,j)}|}{\sum_{x \in A} t_i(1-t_j)|x^{(i,j)}|} - 1 \quad \text{by Lemma 2} \\
&= \frac{|\Phi(T)_0^i - \Phi(T-A)_0^i| - |A|}{|A|} \\
&= \frac{|\Phi(T)_0^i| - |A \cup \Phi(T-A)_0^i|}{|A|} \quad \text{as before} \\
&= \frac{\partial(|\Phi(T)_0^i| - |A \cup \Phi(T-A)_0^i|)/\partial t_i}{\partial |A|/\partial t_i} \quad \text{as before} \\
&\leq \frac{\partial |\Phi(T)|/\partial t_i}{\partial |T|/\partial t_i} \quad \text{by the hypothesis of Case 2} \\
&= \frac{\partial |\Phi(T)|/\partial t_j}{\partial |T|/\partial t_j} \quad \text{by the Optimality Condition.}
\end{aligned}$$

Proof of Claim 2. This is based on two lemmas:

Lemma 3.

$$\partial |S|/\partial t_{i,1} = \partial |A|/\partial t_i,$$

$$\partial |\Phi|/\partial t_{i,1} = \partial(|\Phi(T)_0^i| - |A \cup \Phi(T-A)_0^i|)/\partial t_i$$

and

$$\partial |S|/\partial t_{i,1} + \partial |S|/\partial t_{i,2} = \partial |T|/\partial t_i,$$

$$\partial |\Phi|/\partial t_{i,1} + \partial |\Phi|/\partial t_{i,2} = \partial |\Phi|/\partial t_i.$$

Proof. Recall that $t_{i,1} \leq t_i \leq t_{i,2}$. We regard $S = S(t_{i,1}, t_{i,2})$, $t_{i,1} < t_i < t_{i,2}$ as a lower set in

$$\mathcal{T}_2 \times \cdots \times \mathcal{T}_2 \times \mathcal{T}_3 \times \mathcal{T}_2 \times \cdots \times \mathcal{T}_2,$$

whose i th factor is $\mathcal{T}_3$, all the others being $\mathcal{T}_2$, and $T = S(t_i, t_i)$ as a lower set in

$$\mathcal{T}_2 \times \cdots \times \mathcal{T}_2 \times \mathcal{T}_2 \times \mathcal{T}_2 \times \cdots \times \mathcal{T}_2.$$

With this identification $S_0^i = T_0^i$, $S_1^i = (T_0^i - A) + \delta_i$, $S_2^i = T_1^i + \delta_i$ and

$$\partial |S_0^i|/\partial t_{i,1} = |S^{(i)}| = |T^{(i)}| = \partial |T_0^i|/\partial t_i,$$

$$\partial |S_1^i|/\partial t_{i,1} = -\partial |T_0^i - A|/\partial t_i,$$

$$\partial |S_2^i|/\partial t_{i,1} = 0.$$

Then, since S is the disjoint union of S_0^i , S_1^i and S_2^i , we have

$$\begin{aligned}\partial|S|/\partial t_{i,1} &= \partial|S_0^i|/\partial t_{i,1} + \partial|S_1^i|/\partial t_{i,1} + \partial|S_2^i|/\partial t_{i,1} \\ &= \partial|T_0^i|/\partial t_i - \partial|T_0^i - A|/\partial t_i \\ &= \partial|A|/\partial t_i,\end{aligned}$$

which gives the first equation. Similarly,

$$\begin{aligned}\partial|S|/\partial t_{i,2} &= \partial|S_0^i|/\partial t_{i,2} + \partial|S_1^i|/\partial t_{i,2} + \partial|S_2^i|/\partial t_{i,2} \\ &= 0 + \partial|T_0^i - A|/\partial t_i + \partial|T_1^i|/\partial t_i \\ &= \partial|T_0^i - A|/\partial t_i + \partial|T_1^i|/\partial t_i.\end{aligned}$$

Therefore,

$$\begin{aligned}\partial|S|/\partial t_{i,1} + \partial|S|/\partial t_{i,2} &= \partial|T_0^i|/\partial t_i + \partial|T_1^i|/\partial t_i \\ &= \partial(|T_0^i| + |T_1^i|)/\partial t_i = \partial|T|/\partial t_i\end{aligned}$$

justifying the third equation. The proofs of the second and fourth equations are along the same lines but we need the additional observation that for a lower set S of $\mathcal{T}_{q_1} \times \dots \times \mathcal{T}_{q_{i-1}} \times \mathcal{T}_{q_i} \times \mathcal{T}_{q_{i+1}} \times \dots \times \mathcal{T}_{q_d}$,

$$\Phi(S)_a^i = [(S_0^i + a\delta_i) - S_a^i] \cup \Phi(S_a^i \downarrow)_a^i,$$

where $X \downarrow = \{y \in \mathfrak{P} : \exists x \in X \text{ s.t. } y \leq x\}$, the lower set generated by X . Then

$$\begin{aligned}\partial|\Phi(S)|/\partial t_{i,1} &= \partial|\Phi(S)_0^i|/\partial t_{i,1} + \partial|\Phi(S)_1^i|/\partial t_{i,1} + \partial|\Phi(S)_2^i|/\partial t_{i,1} \\ &= \partial|\Phi(S)_0^i|/\partial t_{i,1} + \partial|[(S_0^i + \delta_i) - S_1^i] \cup \Phi(S_1^i \downarrow)_1^i|/\partial t_{i,1} + \partial|\Phi(S)_2^i|/\partial t_{i,1} \\ &= \partial|\Phi(T_0^i)_0^i|/\partial t_i - \partial|A \cup \Phi(T_0^i - A)_0^i|/\partial t_i + 0 \\ &= \partial|\Phi(T)_0^i|/\partial t_i - \partial|A \cup \Phi(T - A)_0^i|/\partial t_i,\end{aligned}$$

which implies the second equation. Also

$$\begin{aligned}\partial|\Phi(S)|/\partial t_{i,2} &= \partial|\Phi(S)_0^i|/\partial t_{i,2} + \partial|\Phi(S)_1^i|/\partial t_{i,2} + \partial|\Phi(S)_2^i|/\partial t_{i,2} \\ &= 0 + \partial|A \cup \Phi(T_0^i - A)_0^i|/\partial t_i + \partial|[(T_0^i + \delta_i) - T_1^i] \cup \Phi(T_1^i \downarrow)_1^i|/\partial t_i.\end{aligned}$$

Therefore,

$$\begin{aligned}\partial|\Phi(S)|/\partial t_{i,1} + \partial|\Phi(S)|/\partial t_{i,2} &= \partial|\Phi(T)_0^i|/\partial t_i + \partial|[(T_0^i + \delta_i) - T_1^i] \cup \Phi(T_1^i \downarrow)_1^i|/\partial t_i \\ &= \partial(|\Phi(T)_0^i| + |[T_0^i + \delta_i) - T_1^i] \cup \Phi(T_1^i \downarrow)_1^i|)/\partial t_i \\ &= \partial(|\Phi(T)_0^i| + |\Phi(T)_1^i|)/\partial t_i = \partial|\Phi(T)|/\partial t_i\end{aligned}$$

giving the fourth equation. $\square$

Lemma 4. If $a, b, c, d > 0$, then $a/b < (a + c)/(b + d) \Leftrightarrow a/b < c/d \Leftrightarrow (a + c)/(b + d) < c/d$.

Proof. $a/b < c/d \Leftrightarrow ad < bc \Leftrightarrow ad + ab < bc + ab \Leftrightarrow a(b + d) < b(a + c) \Leftrightarrow a/b < (a + c)/(b + d)$. $\square$

Now, by Lemma 3

$$\frac{\partial|\Phi|/\partial t_{i,1}}{\partial|S|/\partial t_{i,1}} = \frac{\partial(|\Phi(T)_0^i| - |A \cup \Phi(T - A)_0^i|)/\partial t_i}{\partial|A|/\partial t_i}$$

and

$$\frac{\partial|\Phi|/\partial t_{i,1} + \partial|\Phi|/\partial t_{i,2}}{\partial|S|/\partial t_{i,1} + \partial|S|/\partial t_{i,2}} = \frac{\partial|\Phi|/\partial t_i}{\partial|S|/\partial t_i},$$

so the hypothesis of Case 1, combined with Lemma 4, implies Claim 2 and the proof of Theorem 1 is finished.

2.6. Which Hamming balls are actually optimal?

Theorem 2. The unique weighting vector $(t_1, t_2, \dots, t_d) \in [0, 1]^d$ which minimizes

$$|\Phi(HB(r, d)); (t_1, t_2, \dots, t_d)|$$

over all weightings for which

$$|HB(r, d); (t_1, t_2, \dots, t_d)| = v$$

is the constant vector $(\forall i, j \ t_i = t_j)$.

Proof. Suppose not, and that the minimum is achieved at some $(t_1, t_2, \dots, t_d) \in (0, 1)^d$ with $t_i < t_j$ for some i, j . Let

$$HS(r, d) = \left\{ x \in (\mathcal{T}_2)^d : \sum_{i=1}^d x_i = r \right\}$$

be the Hamming sphere, then

$$HB(r, d) = \bigcup_{s=0}^r HS(s, d)$$

and

$$\Phi(HB(r, d)) = HS(r + 1, d).$$

So

$$\partial|HS(r, d); t|/\partial t_i = |HS(r, d - 1); t^{(i)}| - |HS(r - 1, d - 1); t^{(i)}|$$

and

$$\partial|HB(r, d); t|/\partial t_i = \sum_{s=0}^r \partial|HS(s, d); t|/\partial t_i = |HS(r, d - 1); t^{(i)}|$$

from which it follows that

$$\begin{aligned} \frac{\partial |\Phi(HB(r, d)); t| / \partial t_i}{\partial |HB(r, d); t| / \partial t_i} &= \frac{\partial |HS(r + 1, d); t| / \partial t_i}{\partial |HB(r, d); t| / \partial t_i} \\ &= \frac{|HS(r + 1, d - 1); t^{(i)}| - |HS(r, d - 1); t^{(i)}|}{|HS(r, d - 1); t^{(i)}|} \\ &= \frac{|HS(r + 1, d - 1); t^{(i)}|}{|HS(r, d - 1); t^{(i)}|} - 1. \end{aligned}$$

Thus from the optimality condition for lower sets, $\partial |\Phi| / \partial t_i / \partial |S| / \partial t_i = \partial |\Phi| / \partial t_j / \partial |S| / \partial t_j$, we have

$$\frac{|HS(r + 1, d - 1); t^{(i)}|}{|HS(r, d - 1); t^{(i)}|} = \frac{|HS(r + 1, d - 1); t^{(j)}|}{|HS(r, d - 1); t^{(j)}|}$$

which gives

$$\begin{aligned} &\frac{|HS(r + 1, d - 2); t^{(i,j)}| t_j + |HS(r, d - 2); t^{(i,j)}| (1 - t_j)}{|HS(r, d - 2); t^{(i,j)}| t_j + |HS(r - 1, d - 2); t^{(i,j)}| (1 - t_j)} \\ &= \frac{|HS(r + 1, d - 2); t^{(i,j)}| t_i + |HS(r, d - 2); t^{(i,j)}| (1 - t_i)}{|HS(r, d - 2); t^{(i,j)}| t_i + |HS(r - 1, d - 2); t^{(i,j)}| (1 - t_i)}. \end{aligned}$$

So, with $c_r = |HS(r, d - 2); t^{(i,j)}|$ and $g(x) = (c_{r+1}x + c_r(1 - x)) / (c_rx + c_{r-1}(1 - x))$ we have $g(t_i) = g(t_j)$. By Rolle's Theorem then, $\exists t_0, t_i < t_0 < t_j$, such that $dg/dx(t_0) = 0$. However,

$$\frac{dg}{dx}(x) = \frac{c_{r-1}c_{r+1} - c_r^2}{c_rx + c_{r-1}(1 - x)^2} \neq 0$$

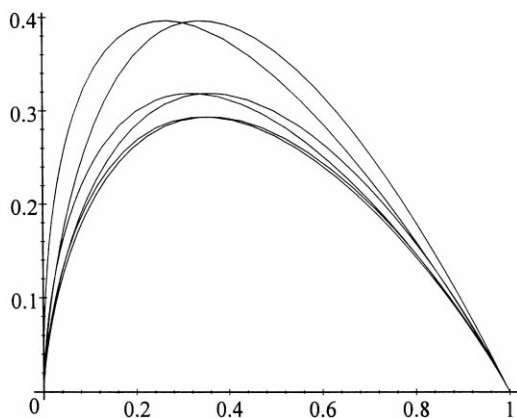
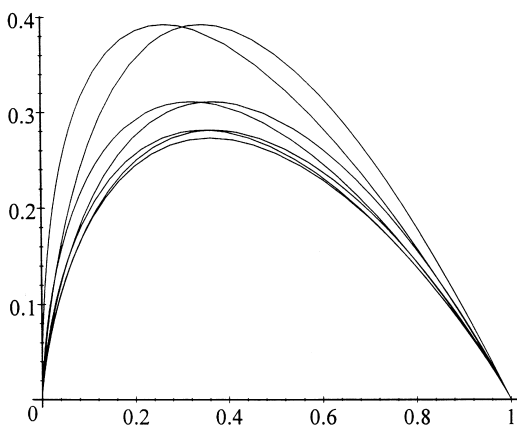
for any x since $c_r^2 > c_{r-1}c_{r+1}$ by Cauchy's Inequality [18] and we have a contradiction. Note that we did not use $t_i > 0$ or $t_j < 1$, so the conclusion applies even if $t_i = 0$ or $t_j = 1$. In those cases the dimension is effectively lowered by one, so the conclusion applies to all $[0, 1]^d$ by induction. $\square$

The dual complement of a Hamming ball is also a Hamming ball:

$$DC(HB(r, d)) = HB(d - r - 2, d).$$

We illustrate the relationship between the functions (of v) $|\Phi(HB(r, d))|$, $r = 0, 1, \dots, d - 2$, with their graphs for $d = 7$ and 8: $|\Phi(r, d; v)|$ is defined parametrically for $0 \leq t \leq 1$ by

$$\begin{aligned} v(r, d; t) &= \sum_{i=0}^r \binom{d}{i} t^{d-i} (1 - t)^i, \\ |\Phi(r, d; t)| &= \binom{d}{r+1} t^{d-r-1} (1 - t)^{r+1}. \end{aligned}$$

Graphs of $|\Phi(r, d; v)|$ for $d=7, 0 \leq r \leq 5$ Graphs of $|\Phi(r, d; v)|$ for $d=8, 0 \leq r \leq 6$

The graphs clearly occur in pairs, with the crossover point of each pair occurring between $\frac{1}{4}$ and $\frac{1}{2}$. Those pairs are actually the dual pairs, $HB(r, d)$ and $HB(d-r-2, d)$, of Hamming balls; the pair for $r=0$ having the highest (w.r.t. $|\Phi|$) crossover point, then $r=1$, etc., for $r \leq (d/2) - 1$. In each pair the ball of lower radius minorizes its mate to the left of their crossover and vice versa to the right. If d is even, $HB((d/2) - 1, d)$ is self-dual. It appears from these graphs that the Hamming balls of radius near $(d/2) - 1$ minorize all the others with respect to vertex-boundary and this is largely true: If

$$v = v(d, r; t) = \sum_{i=0}^r \binom{d}{i} t^{d-i} (1-t)^i = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^R e^{-s^2/2} ds = F(R),$$

then $R = F^{-1}(v)$ is determined by v alone, and $r + 1 \simeq \mu + \sigma R$, where $\mu = d \times t$ and $\sigma = \sqrt{d \times t(1 - t)}$ are mean and variance (of the binomial distribution). Also

$$\begin{aligned} |\Phi(d, r; t)| &= \binom{d}{r + 1} t^{d-r-1} (1 - t)^{r+1} \\ &\simeq \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{(r+1-\mu)/\sigma} e^{-s^2/2} \, ds - \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{(r-\mu)/\sigma} e^{-s^2/2} \, ds \\ &= \frac{1}{\sqrt{2\pi}} \int_{(r-\mu)/\sigma}^{(r+1-\mu)/\sigma} e^{-s^2/2} \, ds \simeq \frac{1}{\sqrt{2\pi}} e^{-R^2/2} / \sigma. \end{aligned}$$

Thus, for a given v , $|\Phi|$ is minimized when σ is maximized, i.e. when $t = \frac{1}{2}$, and then $r + 1 \simeq (d/2) + (\sqrt{d}/2)R$. For the great preponderance of $v \in [0, 1]$ the value of $\sqrt{d}F^{-1}(v)$ will be much smaller than d .

On the other hand, it is not entirely true since

$$\lim_{v \rightarrow 0} \frac{|\Phi(d, r; v)|}{|\Phi(d, r + 1; t)|} = 0$$

for all r , $0 \leq r \leq d - 2$, as may be shown by L'Hopital's Rule. So $HB(d, 0)$ is optimal for v sufficiently small and $DC(HB(d, 0)) = HB(d, d - 2)$ is optimal for v near 1. It seems likely that every Hamming ball will be optimal for some interval of v 's, but for the Hamming balls with r not near $(d/2) - 1$, the interval of optimality will be insignificant, as will the relative value of $|\Phi|$ there.

3. Extensions and variations

3.1. Exact solutions for certain values of ℓ

In Section 1 we introduced the vertex-isoperimetric problem on the Hamming graph $(K_q)^d$ and in Section 2 we reduced that to minimizing $|\Phi(S)|$ over all lower sets, S , of cardinality ℓ in $(\mathcal{T}_q)^d$. Then we passed to the continuous limit, letting $q \rightarrow \infty$, and got the problem of minimizing the Lebesgue measure of $\Phi(S)$ over all lower sets in $[0, 1]^d$ of Lebesgue measure v . Finally, we reduced that to minimizing $|\Phi(T); t|$ over all lower sets, T , of $\mathcal{T}_{q_1} \times \mathcal{T}_{q_2} \times \cdots \times \mathcal{T}_{q_d}$ with $|T; t| = v$ and to $q_i = 2$ in particular. Note however, that an instance of the original problem may be embedded into the last one with $q_i = q$ for $i = 1, \dots, d$: Let the weight of T be given by $t_{i,j} = (j/q)|T; t|$ is then just $|T|/q^d$ and $|\Phi(T); t| = |\Phi(T)|/q^d$. So if the solution of the Continuous Urbanke Problem for $v = \ell/q^d$ corresponds to such a T , then T is a solution to the original discrete problem. There are actually quite a few such solutions, an asymptotically dense set of them. We exemplify this for $d = 8$ and $q = 20$: In that case there are seven possible values for r ranging from 0 to 6, and 19 possible values of r ranging from 1 to 19. That makes $7 \times 19 = 133$ possibilities altogether, of which 25 are actually solutions. These

are listed below by increasing order of $\ell = \sum_{s=0}^r \binom{8}{s} j^{8-s}(20-j)^s$. Values of r, j and $|\Phi| = \binom{8}{r+1} j^{8-r-1}(20-j)^{r+1}$ are also listed in subsequent columns.

ℓ	r	j	$ \Phi $
153	1	1	1018
18688	1	2	580608
303993	1	3	5899068
6203061	2	3	66856104
31522816	2	4	234881024
108203125	2	5	590625000
289080576	2	6	1194891264
648128341	2	7	2067798824
1275068416	2	8	3170893824
2715927165	3	7	4800247270
4445962240	3	8	5945425920
6665746365	3	9	6724172070
9300000000	3	10	7000000000
12210081565	3	11	6724172070
15208611840	3	12	5945425920
18083825565	3	13	4800247270
21154037760	4	12	3170893824
22884072835	4	13	2067798824
24116028160	4	14	1194891264
24901171875	4	15	590625000
25333596160	4	16	234881024
25526940835	4	17	66856104
25593796939	5	17	5899068
25559400704	5	18	580608
25559989739	5	19	1018

Remark 1. We know what the optimal lower set for $\ell = 1$ is and it corresponds to $r = 0, j = 1$. It is not on this list however, because it is not locally optimal, i.e. optimal for the continuous problem. We have observed that it is locally optimal for v sufficiently small but $v = (\frac{1}{20})^8 \simeq 4 \times 10^{-11}$ is just not small enough for $d = 8$. If we had taken $q = 100$ so that $v = (\frac{1}{100})^8 = 10^{-16}$, then our optimal set with $\ell = 1$ would also be locally optimal.

3.2. Hamming distances greater than 1

For a pair of vertices x, y in $(K_q)^d$ the *Hamming distance* between them is defined to be

$$H(x, y) = |\{i: x_i \neq y_i\}|.$$

The edges of the Hamming graph correspond to those pairs at Hamming distance 1. This suggests extending Urbanke's Problem to graphs with the same vertices but having

an edge between any pair of vertices x, y with $H(x, y) \leq h$ for fixed h , $0 < h < d$. This extension of Urbanke’s Problem is easily solved in the case $q = 2$: Because those problems have Moghadam’s Properties A and B (see Appendix B) the sets which solve the problem for $h = 1$ solve it for all h . However, as we observed in Section 2.1, Urbanke’s Problem does not have Properties A and B for $q > 2$. We can however assert the following:

Theorem 3. $\forall v, 0 < v < 1, \exists r$ such that $HB(r, d)$ minimizes

$$|\Phi_h(r, d)| = |\{y \notin S: \exists x \in S, H(x, y) \leq h\}|$$

over all lower sets, $S \subseteq [0, 1]^d$, with $|S| = v$. The optimal weighting for $HB(r, d)$ is constant.

We leave the verification of this as an exercise for the reader. Most of our proof for $h = 1$ does not depend on h or is very easily extended. Lemma 2, in the proof of Claim 1, does require some attention but follows from the argument used in [15] to extend stabilization to the h -*path* of a graph, G (the graph with the same vertex set and edges between all pairs connected by a path in G of length h or less). Note however that the interval of optimality for $HB(r, d)$ does depend on h . It seems that the division points for $h + 1$ will interleave those for h .

3.3. Related unsolved problems

There are many unsolved problems to which the methods of this paper can probably be applied. We shall only describe four here.

3.3.1. Sapozhenko’s problem

Already mentioned in Leader’s survey [17] it was brought to our attention by A. Sapozhenko as useful in extending the work which he and Korshynov have done on Dedekind’s Problem. It is the vertex-analog of the Kleitman–West Problem. The graph for the Kleitman–West Problem (and Sapozhenko’s Problem) has as vertices all n -tuples of 0’s and 1’s with exactly k 1’s. There is an edge between a pair of vertices if they differ in exactly two places.

3.3.2. Gierz’s scheduling problem

Let $\mathcal{P}$ be a finite poset which can be thought of as representing the precedence constraints for parts of a project. A one-to-one order-preserving function $\eta : \mathcal{P} \rightarrow \mathcal{T}_p$, where $p = |\mathcal{P}|$, is called a *schedule*. The *bandwidth-scheduling problem* then is to minimize

$$\max\{\eta(y) - \eta(x): y \text{ covers } x \text{ in } \mathcal{P}\}.$$

Moghadam [21] and Bollobas-Leader [2] were able to solve the bandwidth-scheduling problem for $\mathcal{P} = (\mathcal{T}_q)^d$ because it has Properties A and B (see Appendix B). My

colleague, G. Gierz, has suggested [9] that the bandwidth-scheduling problem for

$$\mathcal{K}_{d,r} = \left\{ x \in (\mathbb{Z}_+)^d : \sum_{i=1}^d x_i \leq r \right\}$$

is interesting. Even though the balls which Chvatalova et al. found to be optimal (i.e. minimizing $|\Phi(S)|$ over all lower sets S in $(\mathcal{T}_q)^d$) are subsets of $\mathcal{K}_{d,r}$, they are not the only locally optimal sets and not even optimal among those. We conjecture that the sets

$$S(d, r', t) = \left\{ x \in (\mathbb{Z}_+)^d : \sum_{i=1}^t x_i \leq r' \right\}$$

constitute all the locally optimal lower sets of $\mathcal{K}_{d,r}$ (up to symmetry) and that the optimal schedule has initial segments of the form $S(d, r', \lfloor d/2 \rfloor)$.

3.3.3. The Doctor's Waiting Room Problem (DWRP)

The DWRP is studied in a paper on queueing by Foschini and Gopinath [6]: Given $w_1, w_2, \dots, w_d > 0$ and $C_1, C_2, \dots, C_d > 0$, for $x \in \mathcal{K}_{d,r}$ let

$$W(x) = \sum_{i=1}^d C_i w_i^{x_i}$$

and for $A \subseteq \mathcal{K}_{d,r}$ let

$$W(A) = \sum_{x \in A} W(x).$$

Also let

$$A_i = \{x \in A : x_i = 0\}.$$

Then the DWRP is to minimize the ratio $\sum_{i=1}^d W(A_i)/W(A)$ over all lower sets in $\mathcal{K}_{d,r}$. Foschini and Gopinath conjecture that there exists a function $k : \{0, 1\}^d \rightarrow \mathbb{R}_+$ such that

$$A(k; d) = \left\{ x \in \mathcal{K}_{d,r} : \sum_{i=1}^d x_i y_i \leq k(x) \quad \forall y \in \{0, 1\}^d \right\}$$

is a solution to the DWRP.

3.3.4. A classical isoperimetric problem

In [4] it is mentioned that the isoperimetric problem in the real projective plane is unsolved in the sense that there is no proof of the obvious conjecture: For small values, v , of the area the solutions must be spherical caps since the sphere is locally Euclidean. For v near 1 (the area of the whole plane) a solution must be the complement of one for small v , i.e. a band about the equator. These two, which are the only apparent candidates, cross over at $v = \frac{1}{2}$. If these are indeed the solutions, then they are not nested, so Steiner symmetrization, which is monotonic ($A \subseteq B$ imply $\mathcal{S}(A) \subseteq \mathcal{S}(B)$),

cannot give a proof. What would be needed then for a Steiner-type proof to work, would be a nonmonotonic (i.e. discontinuous) variant of Steiner symmetrization. Is it possible that our solution of Urbanke’s Problem, based on a nonmonotonic analog of Steiner symmetrization, could show the way to such a proof? In our monograph [15] where we seek to develop a theory of Steiner operations for graphs and other discrete structures, we included monotonicity as one of the defining properties because it seemed to be necessary to ensure convergence and was possessed by all examples that we knew of (Leader has since pointed out that the argument of Bollobas and Leader [3] involves nonmonotonic operations). Clearly, we have a need for more theoretical insight into the nature of these nonmonotonic “Steiner operations”, so that we may take advantage of the broader opportunities for applications.

Acknowledgements

We are grateful to Sergei L. Bezrukov of the University of Paderborn, Germany, for his careful reading of preliminary versions of this paper. His helpful suggestions and constructive criticisms have made it a much better paper. Any remaining mistakes or obscurities are the author’s responsibility.

Appendix A. A Summary of the theory of stabilization for Φ (see [16])

If $G = (V, E, \partial)$ is a graph embedded in $\mathbb{R}^d$, d -dimensional Euclidean space, and $\mathfrak{R}$ is a reflection which acts as a symmetry of G , then

Definition A.1. $\mathfrak{R}$ is called stabilizing if $\forall e \in E$, if $\partial(e) = \{x, y\}$ and x, y are on opposite sides of the fixed hyperplane of $\mathfrak{R}$, then $\mathfrak{R}(x) = y$.

Definition A.2. If $\mathfrak{R}$ is stabilizing for G , $p \in \mathbb{R}^d$ is not fixed by $\mathfrak{R}$ and $S \subseteq V$ with

$$\Sigma_0 = \{x \in S : ||x - p|| \leq ||\mathfrak{R}(x) - p||\}, \tag{A.1}$$

$$\Sigma_1 = \{x \in S : ||x - p|| > ||\mathfrak{R}(x) - p|| \ \& \ \mathfrak{R}(x) \in S\}, \tag{A.2}$$

$$\Sigma_2 = \{x \in S : ||x - p|| > ||\mathfrak{R}(x) - p|| \ \& \ \mathfrak{R}(x) \notin S\}, \tag{A.3}$$

then

$$Stab_{\mathfrak{R},p}(S) = \Sigma_0 \cup \Sigma_1 \cup \mathfrak{R}(\Sigma_2).$$

Theorem A.1. *The operator $Stab_{\mathfrak{R},p}$ has the following properties:*

1. $|Stab_{\mathfrak{R},p}(S)| = |S|$,
2. $|\Phi(Stab_{\mathfrak{R},p}(S))| \leq |\Phi(S)|$ and
3. If $S \subseteq T$, then $Stab_{\mathfrak{R},p}(S) \subseteq Stab_{\mathfrak{R},p}(T)$.

Eq. (A.1) follows directly from the definition of $Stab_{\mathfrak{R},p}(S)$. Inequality (A.2) is based upon the observation that for any vertex in $\Phi(Stab_{\mathfrak{R},p}(S))$ but not in $\Phi(S)$, there is a corresponding vertex (its image under $\mathfrak{R}$) which is in $\Phi(S)$ but not in $\Phi(Stab_{\mathfrak{R},p}(S))$. There are two ways that this can happen depending on which side of the fixed hyperplane the vertex lies.

Given G and stabilizing reflections $\mathfrak{R}_0, \mathfrak{R}_1, \dots, \mathfrak{R}_{k-1}$ and $p \in \mathbb{R}^d$ not fixed by any $\mathfrak{R}_i$, then define a sequence of operators $\mathcal{S}_j : 2^V \rightarrow 2^V$, $j = 0, 1, \dots$ by $\mathcal{S}_0(S) = S$ (i.e. $\mathcal{S}_0$ is the identity) and $\mathcal{S}_{j+1} = Stab_{\mathfrak{R}_{j(\bmod k)},p} \circ \mathcal{S}_j$.

Theorem A.2. $\exists j_0 \in \mathbb{Z}_+$ such that $\forall j \geq j_0$ and $\forall S \subseteq V$, $\mathcal{S}_{j+1}(S) = \mathcal{S}_j(S)$, i.e.

$$Stab_{\mathfrak{R}_{j(\bmod k)},p}(\mathcal{S}_j(S)) = \mathcal{S}_j(S).$$

Definition A.3. A set $S \subseteq V$ such that $Stab_{\mathfrak{R}_i,p}(S) = S$ for $i = 0, 1, \dots, k-1$, is called stable.

Theorems A.1 and A.2 show that in minimizing $|\Phi(S)|$ over 2^V we need only consider stable sets. But how can we tell which sets are stable?

Definition A.4. Let $\mathfrak{S}(\mathfrak{R}_i; p) = \{(x, y) : \mathfrak{R}_i(x) = y \text{ \& } \|x - p\| < \|y - p\|\}$. Then the stability order,

$$\mathfrak{S}(\mathfrak{R}_0, \mathfrak{R}_1, \dots, \mathfrak{R}_{k-1}; p)$$

is the reflexive transitive closure of $\bigcup_{i=0}^{k-1} \mathfrak{S}(\mathfrak{R}_i; p)$.

Remark A.1. If $\mathfrak{R}_0, \mathfrak{R}_1, \dots, \mathfrak{R}_{k-1}$ constitute all of the reflections of a Coxeter group then $\mathfrak{S}(\mathfrak{R}_0, \mathfrak{R}_1, \dots, \mathfrak{R}_{k-1}; p)$ is the Bruhat order on a quotient of that group. If they are just the basic reflections (in the sense of Coxeter), then it is a weak Bruhat order.

Definition A.5. A subset S of a poset $\mathcal{P}$, is called a lower set (also downset or ideal) if $x < y$ and $y \in S$ imply that $x \in S$.

Theorem A.3. A set $S \subseteq V$ is stable iff it is a lower set in $\mathfrak{S}(\mathfrak{R}_0, \mathfrak{R}_1, \dots, \mathfrak{R}_{k-1}; p)$.

Appendix B. A summary of the theory of compression for Φ

Definition B.1. A graph, $G = (V, E, \partial)$, is said to have Property A if there exists a numbering, $\eta_0 : V \rightarrow \{1, \dots, |V|\}$, one-to-one and onto, such that $\forall \ell$, $S_\ell = \eta_0^{-1}\{1, \dots, \ell\}$ minimizes $|\Phi(S)|$ over all $S \subseteq V$ with $|S| = \ell$. Since $\emptyset = S_0 \subset S_1 \subset \dots \subset S_{|V|}$ this has also been called the nested solutions property.

Definition B.2. A graph, G , having Property A is said to have Property B if

$$S_{\ell+|\Phi(S_\ell)|} = S_\ell \cup |\Phi(S_\ell)|.$$

Definition B.3. If $H = (V_H, E_H, \partial_H)$ has Property A and $J = (V_J, E_J, \partial_J)$ is any graph, then we define the compression operator for their product

$$H \times J = (V_H \times V_J, E_H \times V_J \cup V_H \times E_J, \partial_H \times V_J \cup V_H \times \partial_J)$$

by

$$\mathcal{Comp}(S) = \bigcup_{w \in V_J} S_{\ell_w} \times \{w\},$$

where $S \subseteq V_{H \times J}$ and $\ell_w = |S \cap (V_H \times \{w\})|$.

Theorem B.1. *If H has property B as well as property A,*

1. $|\mathcal{Comp}(S)| = |S|$,
2. $|\Phi(\mathcal{Comp}(S))| \leq |\Phi(S)|$ and
3. $S \subseteq T$ implies $\mathcal{Comp}(S) \subseteq \mathcal{Comp}(T)$.

Given G and a numbering, η , of G of G which is consistent on $H_i \times \{w\}$ for each of the factorizations $H_0 \times J_0, H_1 \times J_1, \dots, H_{k-1} \times J_{k-1}$ of G which give compressions $\mathcal{Comp}_0, \mathcal{Comp}_1, \dots, \mathcal{Comp}_{k-1}$, define a sequence of operators $\mathcal{C}_0, \mathcal{C}_1, \dots$ by $\mathcal{C}_0(S) = S$ and $\mathcal{C}_{j+1} = \mathcal{Comp}_{j \pmod k} \circ \mathcal{C}_j$.

Theorem B.2. $\exists j_0 \in \mathbb{Z}_+$ such that $\forall j \geq j_0 \mathcal{C}_{j+1}(S) = \mathcal{C}_j(S) \ \forall S \subseteq V$, i.e.

$$\mathcal{Comp}_{j \pmod k}(\mathcal{C}_j(S)) = \mathcal{C}_j(S).$$

Definition B.4. A set $S \subseteq V$ such that $\mathcal{Comp}_i(S) = S$ for $i = 0, 1, \dots, k - 1$, is called compressed.

Theorems B.1 and B.2 show that in minimizing $|\Phi(S)|$ over all S in 2^{V_G} we need only consider compressed sets. But how can we tell which sets are compressed?

Definition B.5. Let

$$\mathfrak{C}(H_i \times J_i) = \{(x, y) \colon x, y \in H_i \times \{w\}, \ w \in J_i \text{ and } \eta_i(x) < \eta_i(y)\}.$$

Then the compressibility order

$$\mathfrak{C}(H_0 \times J_0, H_1 \times J_1, \dots, H_{k-1} \times J_{k-1}),$$

is the reflexive transitive closure of $\bigcup_{i=0}^{k-1} \mathfrak{C}(H_i \times J_i)$.

Theorem B.3. *A set $S \subseteq V$ is compressed iff it is a lower set in the compressibility order.*

Remark B.1. Both compression and stabilization are discrete analogs of Steiner symmetrization [20].

Appendix C. On the variational method

One of the classic strategies for “solving” a discrete problem is to

1. pass to a continuous limit in such a way as to approximate, or somehow retain the essence of, the original problem,
2. use analysis to find the solution of the continuous problem, and then
3. draw conclusions about the original, discrete problem from it.

This technique is so powerful that combinatorial methods could not compete, except when wielded by a master of computation such as Leonhard Euler. As a result, combinatorics grew very little and was not taken seriously by most mathematicians until about 1960. Then the digital electronic computer opened up a whole new world because it could actually carry out the myriads of steps in calculations which had previously only been possible “in principle”. Computer science has also provided new discrete problems and raised the ante for solving them. As the problems and methods become more sophisticated, even the limits of the computer’s abilities are often surpassed. This paper is one of a series [13,14] in which the author has explored this synergy between combinatorics and analysis.

When one passes to a continuous limit with a combinatorial optimization problem, the result is often infinite-dimensional and that is the case in this paper. There is a theory for such problems if the objective function and constraints can be given integral representations; the Euler–Lagrange calculus of variations [7], but often it is not clear how to do this or even if it is possible. Essentially, all the successes for this approach to date have been derived by ad hoc and relatively elementary means. One of these seemingly “accidental” properties, which somehow keep coming up, is that the infinite-dimensional problem reduces to a finite-dimensional one. The problem can then be handled by multivariate calculus and that is what happens here.

For the isoperimetric problem of Section 2 we propose to perturb a lower set, $S = \phi^{-1}(T)$, T being a lower set of $\mathcal{T}_{q_1} \times \mathcal{T}_{q_2} \times \cdots \times \mathcal{T}_{q_d}$, having weight $t = (t_{i,j})$, elevating the $t_{i,j}$ facet by a small amount, ε , and depressing the $t_{k,l}$ facet by $-f(\varepsilon)$ so as to maintain a constant volume. The objective function may then be represented as a function of ε ,

$$F(\varepsilon) = |\Phi(S(t_{i,j} + \varepsilon, t_{k,l} + f(\varepsilon)))|,$$

where $f(\varepsilon)$ is defined implicitly by $|S(t_{i,j} + \varepsilon, t_{k,l} + f(\varepsilon))| = v = |S(t_{i,j}, t_{k,l})|$. Since $|\Phi|$ and $|S|$ are polynomials in the $t_{i,j}$ ’s, f and F have all necessary derivatives near $\varepsilon = 0$. Then

$$v'(0) = \frac{\partial |S(t_{i,j}, t_{k,l})|}{\partial t_{i,j}} + \frac{\partial |S(t_{i,j}, t_{k,l})|}{\partial t_{k,l}} f'(0)$$

since $f(0) = 0$. But $v'(0) = 0$ also and we may solve for $f'(0)$

$$f'(0) = - \frac{\partial |S(t_{i,j}, t_{k,l})| / \partial t_{i,j}}{\partial |S(t_{i,j}, t_{k,l})| / \partial t_{k,l}}.$$

Similarly,

$$\begin{aligned} F'(0) &= \frac{\partial |\Phi(t_{i,j}, t_{k,l})|}{\partial t_{i,j}} + \frac{\partial |\Phi(t_{i,j}, t_{k,l})|}{\partial t_{k,l}} f'(0) \\ &= \frac{\partial |\Phi(t_{i,j}, t_{k,l})|}{\partial t_{i,j}} - \frac{\partial |\Phi(t_{i,j}, t_{k,l})|}{\partial t_{k,l}} \frac{\partial |S(t_{i,j}, t_{k,l})| / \partial t_{i,j}}{\partial |S(t_{i,j}, t_{k,l})| / \partial t_{k,l}}. \end{aligned}$$

$F''(0)$ may also be computed in the same way and we have by Taylor's Theorem that

$$\begin{aligned} |\Phi(S(t_{i,j} + \varepsilon, t_{k,l} + f(\varepsilon)))| &= |\Phi(S(t_{i,j}, t_{k,l}))| + \left\{ \frac{\partial |\Phi|}{\partial t_{i,j}} \frac{\partial |S|}{\partial t_{k,l}} - \frac{\partial |\Phi|}{\partial t_{k,l}} \frac{\partial |S|}{\partial t_{i,j}} \right\} \frac{\varepsilon}{\partial |S| / \partial t_{k,l}} \\ &\quad + \left\{ \frac{\partial |\Phi|}{\partial t_{k,l}} \left(-\frac{\partial^2 |S|}{\partial t_{i,j}^2} \left(\frac{\partial |S|}{\partial t_{k,l}} \right)^2 + 2 \frac{\partial^2 |S|}{\partial t_{i,j} \partial t_{k,l}} \frac{\partial |S|}{\partial t_{i,j}} \frac{\partial |S|}{\partial t_{k,l}} \right. \right. \\ &\quad \left. \left. - \frac{\partial^2 |S|}{\partial t_{k,l}^2} \left(\frac{\partial |S|}{\partial t_{i,j}} \right)^2 \right) + \frac{\partial^2 |\Phi|}{\partial t_{i,j}^2} \left(\frac{\partial |S|}{\partial t_{i,j}} \right)^3 \right. \\ &\quad \left. - 2 \frac{\partial^2 |\Phi|}{\partial t_{i,j} \partial t_{k,l}} \frac{\partial |S|}{\partial t_{i,j}} \left(\frac{\partial |S|}{\partial t_{k,l}} \right)^2 + \frac{\partial^2 |\Phi|}{\partial t_{k,l}^2} \left(\frac{\partial |S|}{\partial t_{i,j}} \right)^2 \frac{\partial |S|}{\partial t_{k,l}} \right\} \\ &\quad \times \frac{\varepsilon^2}{2 \left(\frac{\partial |S|}{\partial t_{k,l}} \right)^3} + O(\varepsilon^3). \end{aligned}$$

References

- [1] B. Bollobas, I. Leader, Edge-isoperimetric inequalities in the grid, *Combinatorica* 11 (1991) 299–314.
- [2] B. Bollobas, I. Leader, Compressions and isoperimetric inequalities, *J. Combin. Theory Ser. A* 56 (1991) 47–62.
- [3] B. Bollobas, I. Leader, Isoperimetric inequalities and fractional set systems, *J. Combin. Theory Ser. A* 56 (1991) 63–74.
- [4] Y.D. Burago, V.A. Zalgaller, *Geometric inequalities*, Grundlehren Math. Wiss., vol. 285, Springer, Berlin, 1980.
- [5] J. Chvatalova, Optimal labeling of a product of two paths, *Discrete Math.* 2 (1975) 249–253.
- [6] G.J. Foschini, B. Gopinath, Sharing memory optimally, preprint, Bell Labs, Murray Hill, NJ.
- [7] C. Fox, *An Introduction to the Calculus of Variations*, Dover, New York, 1950.
- [8] M.R. Garey, D.S. Johnson, *Computers and Intractability: A Guide to the Theory of NP-Completeness*, W.H. Freeman & Co., San Francisco, 1979.
- [9] G. Gierz, private communication, c. 1987.
- [10] P.R. Halmos, *Measure Theory*, van Nostrand, Princeton, 1950.
- [11] L.H. Harper, Optimal assignments of numbers to vertices, *J. SIAM* 12 (1964) 131–135.
- [12] L.H. Harper, Optimal numberings and isoperimetric problems on graphs, *J. Combin. Theory* 1 (1966) 385–393.
- [13] L.H. Harper, A Continuous analog of Sperner's theorem, *Pacific J. Math.* 118 (1985) 411–425.
- [14] L.H. Harper, On a problem of Kleitman and West, *Discrete Math* 93 (1991) 169–182.
- [15] L.H. Harper, Global methods of combinatorial optimization, preprint.
- [16] S. Karlin, *Total Positivity*, Stanford Univ. Press, Stanford, CA, 1968.
- [17] I. Leader, Discrete isoperimetric inequalities, *Proc. Symp. Appl. Math.*, vol. 44 (Probabilistic Combinatorics and its Applications), 1991, pp. 57–80.

- [18] J.H. Lindsey II, Assignments of numbers to vertices, *Amer. Math. Monthly* 71 (1964) 508–516.
- [19] H.S. Moghadam, Compression operators and a solution to the bandwidth problem of the product of n paths, Ph. D. Thesis, Univ. of Calif., Riverside, 1983.
- [20] G. Polya, G. Szego, Isoperimetric Inequalities in Mathematical Physics, *Annals of Math. Studies*, vol. 27, Princeton Univ. Press, Princeton, 1951.
- [21] R. Urbanke, Private communication, December 1996.